

Zhang

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Education

San Jose State University

2024/08 - Present

M.S in Software Engineering

The Pennsylvania State University

2018/08–2023/05

B.S in Computational Mathematics, Minor in Computer Science

Related Courses: Network and Telecommunication, Intermediate Programming, Object-Oriented Programming with web, Data Structure and Algorithm, Cloud Technologies, Enterprise Software Platform

Work Experience

Ma Labs | *IT Networking Intern* | On-site

2023/09–2024/06

- Developed robust C# scripts to automate mailbox creation, transferring user data to MS Exchange and MySQL databases, enhancing data storage and retrieval efficiency
- Designed and implemented advanced features to simplify user authorization and authentication
- Worked with cross-functional teams to optimize onboarding workflows and account management process

Penguin Random House LLC. | *Contractor* | Remote

2023/04–2023/08

- Designed a chatbot prototype leveraging GPT-3.5 for semantic analysis of authors' works, utilizing Python-based AI workflows, rapid prototyping, and LangChain integration to enhance pipeline flexibility
- Engineered interactive Q&A features to enable user queries on analyzed text content

Project Experience

Application Development | CMPE 272 | San Jose State University

2024/12

- Led a four-person team in developing a deep learning-based pneumonia detection application using a fine-tuned VGG19 model in Python, achieving 92% test accuracy
- Employed Data Augmentation, Dropout, and Early Stopping on Google Colab's T4 GPU for faster training and stronger generalization
- Optimized performance by minimizing test loss to 0.28 and achieving a 0.96 ROC-AUC, leveraging the Adam optimizer, Categorical Crossentropy, and real-time validation monitoring

Numerical Analysis Course Final Project | Penn State University

2023/04

- Led a team of three to solve power iteration problems in MATLAB
- Computed eigenvectors for a PageRank-like algorithm to demonstrate convergence behavior

Factorization and Primality Course Final Project | Penn State University

2021/12

- Developed Python-based algorithms for large integer factorization and primality testing, minimizing reliance on external libraries to ensure efficient solutions
- Applied discrete mathematics principles to optimize complex numeric computations

Skills

- Operating Systems: Linux, Windows Server
- Programming Languages: Python, C#
- Tools: AWS, Visual Studio Code, MS Exchange, MySQL, MATLAB, R studio, Netbeans
- Mathematical skills: Numerical Analysis, Probability, Linear Algebra, Discrete Math